

RAJAN N | Applied AI Engineer

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PROFESSIONAL SUMMARY

Applied AI/ML Engineer with 1.4 years of experience delivering 3 AI systems across enterprise knowledge assistance, business insight generation, and regulatory compliance. Independently owns 1 live application from architecture through deployment and incident resolution, combining model development with API integration and production engineering.

EXPERIENCE

Distinctive India Pvt Ltd | Data Scientist — AI/LLM Engineering Focus | Jul 2025 - Present

- Contributed to delivery of 2 enterprise AI systems at Distinctive, leading development of the Insights platform; ran 12+ ML/LLM experiments, achieving 98% processing accuracy for text insight generation.
- Delivered asynchronous AI services with FastAPI, Docker, Redis, and Celery, integrating inference into queued enterprise workflows.
- Used Vertex AI for ML workflows and experimented with Amazon SageMaker; evaluated AWS Bedrock models across generation, reasoning, and document QA.
- Led Docker/Pytest knowledge-sharing sessions and documented engineering workflows; proposed an AI-first development approach combining human validation with automated reasoning.

PROJECTS

NBFC Compliance Intelligence — Solo, Deployed Live | Aug 2026 - Present
nbfc-compliance-intelligence.nrajan.in

- Owned the loan-document/call-transcript auditor from architecture to deployment, reducing 20-30-minute reviews to sub-minute checks; separated extraction from judgment, applied deterministic rules before LLM assessment, and escalated ambiguity to humans.
- Applied QLoRA fine-tuning and date-aware hybrid retrieval (pgvector, PostgreSQL full-text search, RRF); enforced schema boundaries and code validation so every released verdict cites a real clause effective on the event date.
- Built an evaluation harness tracking extraction accuracy, evidence grounding, citation validity, hallucination rate, and abstention correctness; instrumented per-call token cost and latency through LiteLLM.
- Cut processing latency by ~4x with semaphore-bounded async concurrency; reproduced and resolved Celery/asyncpg event-loop failures and managed-PostgreSQL retrieval incompatibilities.

DataSymphony Knowledge Companion — Distinctive India Pvt Ltd | Mar - Jun 2026

- Built multi-agent orchestration and state-machine workflows for document retrieval, SQL generation, and Python-generated visualizations; implemented a multi-cloud layer with task-specific model routing.
- Collaborated on continuous document ingestion, chunking, retrieval, and reranking; benchmarked FAISS/Pinecone/pgvector using MRR and implemented asynchronous processing, caching, and streaming for real-time queries.
- Versioned prompts and benchmarked models across task-specific test sets, improving accuracy by 14%, reducing cost by 30%, and lowering latency by 10%.
- Automated evaluation cycles and security testing covering ABAC, row-level security (RLS), regulatory requirements, and audit trails to validate enterprise access boundaries and agent behavior.

AI-Powered DPO Insights Generation System — Distinctive India Pvt Ltd | Jan - Feb 2026

- Built a multi-label categorization pipeline from data collection and NLP preprocessing through sentence embeddings, category-sample similarity scoring, and XGBoost; tuned learning rates and engineered aggregate features, reducing feature dimensionality by ~35% while assessing the accuracy-cost tradeoff.
- Benchmarked inexpensive models and implemented confidence-based LLM routing for weak ML predictions; applied QLoRA fine-tuning to insight generation, targeting ~30% lower generation cost against the baseline.
- Developed continuous-learning MLOps with model-drift monitoring, human feedback, and LLM-reviewed corrections; fed validated examples into retraining and ML CI/CD evaluation cycles using 5-fold weighted-F1 cross-validation.
- Batched category-based insight generation to reach ~\$0.0001 per record; optimized classification, fallback, and generation latency, with model artifacts in S3 and inference configurations in PostgreSQL RDS.

LLM Safety Evaluation — Independent Security Research | Jul 2026

- Adversarially tested DeepSeek's conversational LLM for prompt-injection resistance and instruction-hierarchy enforcement, identified a vulnerability, and submitted a responsible disclosure report to the DeepSeek team.

KEY SKILLS

Async Python, Machine Learning, XGBoost, NLP, Feature Engineering, Sentence Embeddings, Multi-Label Classification, MLOps, ML CI/CD, Model-Drift Monitoring, Continuous Learning, LoRA/QLoRA Fine-Tuning, Quantization, GPT/Claude/Gemini, Multi-Model Orchestration, Multi-Agent Orchestration, State-Machine Design, RAG Architecture, Document Ingestion, Chunking Strategies, Retrieval & Reranking, MRR, FAISS/Pinecone/pgvector, LangChain, Prompt Engineering & Versioning, LLM Benchmarking, LLMops, Guardrails, Audit Trails, ABAC/RLS, Docker, Kubernetes (learning), Serverless Deployment, Redis, Celery, Airflow, FastAPI, AWS Bedrock/EC2/S3/RDS, PostgreSQL, Vertex AI, SageMaker, Integration & Security Testing, Production Deployment & Monitoring

EDUCATION

Bachelor of Technology (Information Technology) | 2021 - 2025
Global Institute of Engineering and Technology (GIET), Vellore, Tamil Nadu, India

CERTIFICATIONS

Certification in **Data Science & Machine Learning** | May 2024 - Feb 2025
Illinois Institute of Technology, United States | Entri Elevate | National Skill Development Corporation

Certification in **AI Engineer Core Track: LLM Engineering, RAG, QLoRA, Agents** | Apr 2026 - Present
Ed Donner at Udemy